5 AI-Powered Urban Air Quality Intelligence for Smart City Intervention

Theme: *Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health*

PROBLEM CONTEXT

India's air quality crisis is not a Delhi problem — it is a national urban crisis. In 2024-25, Delhi averaged an AQI of 218 (classified 'Poor' or worse for over 200 days), but the situation across other metros is nearly as severe: Mumbai recorded dangerous AQI levels on over 60 days in 2024, Kolkata averaged AQI above 150 for large parts of the winter season, and Bengaluru and Chennai — long considered relatively clean cities — have seen measurable deterioration as vehicle density and construction activity surge. CPCB's National Air Quality data for 2024 shows that 24 of India's 50 most polluted cities are Tier 1 or Tier 2 urban centres. The Lancet Planetary Health journal estimated 1.67 million premature deaths annually from air pollution in India — a public health burden that falls disproportionately on urban populations. Despite India deploying over 900 Continuous Ambient Air Quality Monitoring Stations (CAAQMS) under the National Clean Air Programme, a 2024 CAG audit found that only 31% of cities with monitoring data had any actionable multi-agency response protocols linked to those readings. The data exists. The intelligence layer to act on it does not. City administrations need more than dashboards. They need geospatial attribution (which sources are responsible at this location, right now), predictive forecasting (what will AQI be in 24 hours at ward level), and enforcement intelligence (where to deploy inspectors for maximum impact). That combination does not exist today.

CHALLENGE STATEMENT

Build an AI-powered Urban Air Quality Intelligence platform that fuses monitoring station data, satellite imagery, mobility feeds, meteorological forecasts, and geospatial land use layers to move from reactive monitoring to proactive, evidence-based intervention — giving city administrators the tools to reduce pollution at source rather than just measure it.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Geospatial Pollution Source Attribution Engine —** Multi-modal AI agent that analyses spatial-temporal AQI patterns against land use maps, traffic density, construction permits, industrial stacks, and satellite-detected thermal anomalies — attributing pollution by source category at ward or zone level with statistical confidence scores.
- **Hyperlocal Predictive AQI Forecasting Agent —** AI model integrating meteorological forecasts, traffic prediction, seasonal emission calendars, and atmospheric dispersion modelling to provide 24-72 hour AQI forecasts at 1km grid resolution across city boundaries — enabling intervention scheduling rather than reactive advisories.
- **Enforcement Intelligence & Prioritisation Agent —** Agent that correlates pollution hotspot data with registered emission sources — industries, construction sites, waste burning locations, diesel fleet movement — and generates prioritised, evidence-backed enforcement action recommendations for municipal and pollution control authorities with supporting geospatial documentation.
- **Multi-City Comparative Intelligence Dashboard —** Geospatial analytics layer that tracks and compares air quality trends, intervention effectiveness, and compliance metrics across multiple urban centres — enabling city administrators to learn from interventions that worked in comparable cities.

- **Citizen Health Risk Advisory System —** AI platform that generates ward-level health risk alerts, maps population vulnerability (hospitals, schools, outdoor workers, elderly populations) against forecast AQI, and pushes personalised advisories through mobile apps, public displays, and IVR in regional languages — Bengaluru in Kannada, Chennai in Tamil, and so on.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Geospatial Intelligence & Remote Sensing (Sentinel satellite, MODIS)
- Multi-Agent AI Systems
- Real-Time IoT Sensor Data Integration (CAAQMS)
- Atmospheric Dispersion Modelling
- Predictive Analytics
- LLMs for multi-language citizen communication

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Source attribution accuracy versus ground-truth emission inventories, AQI forecast accuracy at hyperlocal resolution (RMSE versus persistence baseline), enforcement recommendation quality rated by domain experts, citizen advisory relevance and language coverage, and demonstrated reduction in response time from signal to intervention.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%